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METHODIST UNIVERSITY GHANA

• MICROBIOLOGY AND INFECTION PREVENTION CONTROL

MNUR 201

END-OF-FIRST SEMESTER EXAMINATION

TIME ALLOWED: 2 HOURS: 30 MINUTES

ANSWER ALL THE QUESTIONS ON THE QUESTION PAPER

LEVEL 200 DAY AND WEEKEND
DANSOMAN AND TEMA CAMPUS

INDEX NUMBER

CIRCLE THE APPROPRIATE ANSWER

- 1) A 26-year-old woman visits her physician because of an unusual vaginal discharge. On examination, the physician observes a thin, homogeneous, white-gray discharge that adheres to the vaginal wall. The pH of the discharge is 5.5 (normal)
 - A. *Corynebacterium* species
 - B. *Staphylococcus epidermidis*
 - C. *Prevotella* species
 - D. *Candida albicans*
 - E. *Lactobacillus* species
- 2) Dental plaque and periodontal disease can be thought of as a continuum of what type of physiological process?
 - A. Biofilm formation
 - B. Normal aging
 - C. Abnormal digestion

- D. Exaggerated immune response
E. Chewing gum
- 3) Which one of the following microorganisms is closely associated with dental caries?
A. *C albicans*
B. *Streptococcus mutans*
C. *P melaninogenica*
D. *Neisseria subflava*
E. *S epidermidis*
- 4) Which of the following bacteria lack a cell wall and are therefore resistant to penicillin?
A. Cyanobacteria
B. Mycoplasmas
C. Bdellovibrios
D. Spirochetes
- 5) A cluster of polar flagella is called
A. lophotrichous
B. amphitrichous
C. monotrichous
D. Peritrichous
- 6) The protein from which the hook and filaments of flagella are composed, is
A. Keratin
B. Flagellin
C. Gelatin
D. Casein
- 7) The cocci which mostly occur in single or pairs are
A. Streptococci
B. Diplococci
C. Tetrads
D. None of these
- 8) Which of the following may contain fimbriae?
A. Gram-positive bacteria
B. Gram-negative bacteria
C. Both (a) and (b)
D. None of these
- 9) Peptidoglycan accounts for _____ of the dry weight of cell wall in many gram positive bacteria
A. 50% or more
B. About 10%
C. 11%+ 0.22%
D. About 20%
- 10) Bacteria having no flagella are unable to
A. Move
B. Reproduce
C. stick to tissue surfaces

- D. grow in nutrient agar
- 11) Which of the following is true about cell wall of gram-positive bacteria?
- A. It consists of multiple layers
 - B. It is thicker than that associated with gram-negative bacteria
 - C. It contains teichoic acids
 - D. All of these
- 12) The purpose of Goosed-necked flasks that Louis Pasteur designed to disprove spontaneous generation is to:
- A. Allow the multiplication of microbes in the broth
 - B. Implicate the role of flies in the development of maggots on rotting meat
 - C. Pasteurize the meat broth
 - D. Prevent air from entering the flask
 - E. Trap the microbes and prevent them from reaching the broth
- 13) Who is known as the father of medical microbiology
- A. Koch
 - B. Pasteur
 - C. Jenner
 - D. Tatum
- 14) Who is known as father of microbiology
- A. Robert hook
 - B. Pasteur
 - C. Leewenhoek
 - D. Jenner
- 15) Who developed smallpox vaccine for the first time
- A. koch
 - B. Jenner
 - C. Lister
 - D. Pasteur
- 16) Which of the following scientists introduced the "Sterilization" technique?
- A. Robert Koch
 - B. Joseph Lister
 - C. John Needham
 - D. Louis Pasteur
- 17) Commensalism
- A. Both organisms benefit from symbiosis
 - B. Neither organism derives benefit or harm
 - C. One organism benefit at the expense of the other
 - D. One organism benefit, the other derives neither benefits nor harm
- 18) Which type of bacterial arrangement is formed when cocci adhere after repeated divisions in one plane?
- A. Tetrads
 - B. Staphylococci
 - C. Streptococci

- D. Diplococci
- 19) Which of the following genera is most commonly associated with long chains of cocci?
- A. Staphylococcus
 - B. Enterococcus
 - C. Bacillus
 - D. Escherichia
- 20) How do the arrangements of Streptococcus differ from Staphylococcus?
- A. Streptococcus forms clusters, while Staphylococcus forms chains.
 - B. Streptococcus forms chains, while Staphylococcus forms clusters.
 - C. Both form chains but in different environmental conditions.
 - D. Both form clusters but in different growth phases.
- 21) How does Staphylococcus divide to form its characteristic arrangement?
- A. In a single plane, forming chains
 - B. In two planes, forming tetrads
 - C. In random planes, forming irregular grape-like clusters
 - D. In a spiral, forming corkscrew shapes
- 22) Which type of bacterial arrangement is formed when cells divide in two planes to produce groups of four?
- A. Clusters
 - B. Tetrads
 - C. Chains
 - D. Cubes
- 23) Which of the following best describes the genus Micrococcus?
- A. Forms irregular clusters due to random cell division
 - B. Divides in two planes to form tetrads
 - C. Forms chains by dividing in a single plane
 - D. Divides in multiple planes to form cubic packets
- 24) How can spirilla be distinguished from spirochetes?
- A. Spirilla are flexible, while spirochetes are rigid.
 - B. Spirilla are rigid, while spirochetes are flexible.
 - C. Spirilla are flat, while spirochetes are spiral.
 - D. Spirilla are pleomorphic, while spirochetes have a single shape.
- 25) What does the term "pleomorphic" refer to in bacterial morphology?
- A. Bacteria that are always spiral-shaped
 - B. Bacteria that exhibit variable shapes and lack a single characteristic form
 - C. Bacteria with a rigid spiral structure
 - D. Bacteria that are exclusively spherical
- 26) What is the primary role of a bacterial capsule in pathogenic bacteria like Streptococcus pneumoniae?
- A. Increasing the bacteria's motility
 - B. Protecting the bacteria from phagocytosis by host immune cells
 - C. Facilitating bacterial spore formation
 - D. Enhancing bacterial DNA replication

- 27) What happens to *Streptococcus pneumoniae* when it lacks a capsule?
- A. It becomes more virulent.
 - B. It resists phagocytosis more effectively.
 - C. It is easily destroyed and loses its ability to cause disease.
 - D. It becomes resistant to desiccation.
- 28) Capsules are often composed of:
- A. Proteins and nucleic acids
 - B. Lipopolysaccharides and fatty acids
 - C. Polysaccharides and sometimes polypeptides
 - D. Phospholipids and sterols
- 29) Which of the following is not a function of the glycocalyx?
- A. Aiding in attachment to surfaces, such as tissues in hosts
 - B. Protecting the cell against desiccation
 - C. Enhancing bacterial motility
 - D. Resisting phagocytosis in pathogenic bacteria
- 30) How does the capsule help bacteria survive in harsh environmental conditions?
- A. By aiding DNA repair
 - B. By containing water and protecting against desiccation
 - C. By facilitating sporulation
 - D. By generating heat-resistant structures
- 31) Why is Antony Van Leeuwenhoek considered the father of bacteriology and protozoology?
- A. He invented the first compound microscope.
 - B. He developed the germ theory of disease.
 - C. He was the first to produce accurate descriptions of bacteria and protozoa.
 - D. He discovered antibiotics to combat bacterial infections.
- 32) What term did Van Leeuwenhoek use to describe the microorganisms he observed?
- A. Microbes
 - B. Animalcules
 - C. Protozoans
 - D. Bacterium
- 33) Which of the following microorganisms was not included in Van Leeuwenhoek's early observations?
- A. Bacteria
 - B. Protozoa
 - C. Viruses
 - D. Vinegar eels
- 34) What is the significance of cross-reactive ("natural") antibodies produced by the microbiota?
- A. They attack the microbiota to maintain balance.
 - B. They help the immune system recognize and fight pathogens.
 - C. They reduce inflammation caused by immune responses.
 - D. They neutralize toxins released by the microbiota.

- 35) How does *Lactobacillus* in the vagina prevent the overgrowth of *Candida albicans*?
- A. producing bacteriocins that kill *Candida albicans*
 - B. fermenting glycogen to maintain an acidic pH
 - C. directly consuming *Candida albicans* cells
 - D. increasing the availability of glycogen
- 36) What is the correct sequence of events in the association between microbes and humans?
- A. Colonization → Contact → Infection
 - B. Contact → Colonization → Infection
 - C. Infection → Contact → Colonization
 - D. Contact → Infection → Colonization
- 37) What is an infectious disease?
- A. A condition where microbes colonize the host without causing harm
 - B. An infection that disrupts or damages tissues and organs
 - C. A change in health caused exclusively by genetic factors
 - D. A mild reaction to dietary changes
- 38) What distinguishes infection from disease?
- A. Infection refers to the presence of microbes, while disease refers to deviation from health.
 - B. Infection is always symptomatic, while disease is always asymptomatic.
 - C. Infection involves microbial death, while disease involves microbial growth.
 - D. Infection and disease are synonymous.
- 39) Which one of the following terms characterizes the interaction between herpes simplex virus and a human?
- A. Parasitism
 - B. Symbiosis
 - C. Endosymbiosis
 - D. Endoparasitism
 - E. Consortia
- 40) Which one of the following agents lacks nucleic acid?
- A. Bacteria
 - B. Viruses
 - C. Viroids
 - D. Prions
 - E. Protozoa
- 41) Which one of the following is a prokaryote?
- A. Bacteria
 - B. Algae
 - C. Protozoa
 - D. Fungi
 - E. Slime molds
- 42) Which one of the following agents simultaneously contains both DNA and RNA?
- A. Bacteria
 - B. Viruses
 - C. Viroids
 - D. Prions

E. Plasmids

43) Which of the following cannot be infected by viruses?

- A. Bacteria
- B. Protozoa
- C. Human cells
- D. Viruses
- E. None of the above

44) Viruses, bacteria, and protists are uniquely characterized by their respective size. True or false?

- A. True
- B. False

45) Quorum sensing in prokaryotes involves

- A. Cell-cell communication
- B. Production of molecules such as acetylated homoserine lactone (AHL)
- C. An example of multicellular behavior
- D. Regulation of genes involved in diverse physiologic processes
- E. All of the above

46) A 16-year-old female patient presented to her family physician with a complaint of an abnormal vaginal discharge and pruritus (itching). The patient denied having sexual activity and recently completed a course of doxycycline for the treatment of her acne. An examination of a Gram-stained vaginal smear revealed the presence of gram-positive, oval cells about 4–8 μm in diameter. Her vaginitis is caused by which of the following agents?

- A. Bacterium
- B. Virus
- C. Protozoa
- D. Fungus
- E. Prion

47) A 65-year-old man develops dementia, progressive over several months, along with ataxia and somnolence. An electroencephalographic pattern shows paroxysms with high voltages and slow waves, suggestive of Creutzfeldt-Jakob disease (CJD). By which of the following agents is this disease caused?

- A. Bacterium
- B. Virus
- C. Viroid
- D. Prion
- E. Plasmid

48) A 22-year-old man presents with a painless 1-cm ulcer on the shaft of his penis. Inguinal lymphadenopathy is present. The patient admits to trading drugs for sex and has several sexual partners. An RPR test result is positive, and syphilis is suspected; however, a Gram stain of a swab specimen from the ulcer shows no bacteria. *Treponema pallidum*, the causative agent of syphilis, cannot be visualized by light microscopy because

- A. It is transparent.
- B. It cannot be stained by ordinary stains.
- C. It has a diameter of less than 0.2 μm
- D. The wavelength of white light is too long.
- E. Rapid movement of the organism prevents visualization.

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- 49) Mycoplasma species lack which of the following components?
- A. Ribosomes
 - B. Plasma membrane
 - C. Both DNA and RNA
 - D. Lipids
 - E. Peptidoglycan
- 50) Archaeobacteria can be distinguished from eubacteria by their lack of
- A. DNA
 - B. RNA
 - C. Ribosomes
 - D. Peptidoglycan
 - E. Nucleus
- 51) What concept did Pasteur's experiments on spontaneous generation help disprove?
- A. The germ theory of disease
 - B. The theory of abiogenesis
 - C. The process of fermentation
 - D. The need for aseptic techniques
- 52) Which of the following processes was demonstrated by Pasteur to be caused by microbes like yeasts and bacteria?
- A. Decomposition of plant tissues
 - B. Fermentation
 - C. Photosynthesis
 - D. Nitrogen fixation
- 53) What was one of Pasteur's major contributions to the field of vaccination?
- A. Discovery of antibiotics for bacterial infections
 - B. Development of vaccines for rabies and anthrax
 - C. Use of antiseptics during surgery
 - D. Invention of the microscope
- 54) What are Neglected Tropical Diseases (NTDs)?
- A. Diseases that affect only tropical plants and animals
 - B. Conditions that thrive among the world's poorest populations and are under-addressed
 - C. Rare genetic conditions confined to tropical regions
 - D. Diseases eradicated in developed countries
- 55) What are emerging diseases?
- A. Diseases that have been eradicated but are now reappearing.
 - B. Newly identified conditions that are increasing in incidence.
 - C. Long-standing diseases that remain constant in prevalence.
 - D. Diseases transmitted exclusively through water.
- 56) Which of the following is an example of an emerging infectious disease that became a pandemic?
- A. Tuberculosis
 - B. HIV/AIDS
 - C. Cholera
 - D. Tetanus